

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Technology (M. Tech)

Semester – I University Examination May 2011

M. Tech. (CAD/CAM) (Theory)

ME – 701 Computer Aided Design

Date: 09.05.2011, Monday

Time: 10:00 a.m. to 1:00 p.m.

Maximum Marks: 70

Instructions:

1. Figures to the right indicate *full* marks.
2. Make suitable assumptions and draw neat figures wherever if required.
3. Use of PSG Design data book is permitted.

SECTION- I

- Q1 (a) Using a simplified description of process of design explain where computational aids might be assistance to the designer. 04
- (b) Explain with example the Boolean operations used for Constructive Solid Geometric Modeling. 03
- (c) State the modelling facilities that are expected in good solid modelling software. 04
- Q2 (a) Explain how does the CAD help to reduce product development cycle? 04
- (b) Explain: "Bresenham's algorithm is more efficient than the DDA line generation algorithm." 02
- (c) Why can't a vector scan display produce as many colours as a raster scan display? 02
- (d) Write an algorithm to scan convert a line having slope more than 45° using DDA algorithm. 04

OR

- Q2 (a) Differentiate between: 02
- 1) Analytic curves and Parametric curves
 - 2) Passive and Interactive graphics
- (b) Explain the working principle of vector scan display devices with the help of neat sketches. 05
- (c) Write Bresenham's circle drawing algorithm with (x_c, y_c) as center and radius r . 05
- Q3 (a) Explain non-parametric form of representing curves. What are the limitations of this form of representation for CAD/CAM systems? 04
- (b) A triangle PQR has its vertices at P (0, 0), Q (4, 0) and R (2, 3). It is to be translated by 4 units in x direction & 2 units in y direction, and then it is to be rotated in anticlockwise direction about the new position of point R through 90 degree. Find the new position of the triangle. 04
- (c) Explain homogeneous coordinate system & differentiate between geometric transformation & coordinates transformation. 04

OR

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- Q3 (a)** Explain the curves having following conditions: 04
 1) First order continuity
 2) Second order continuity
- (b)** Distinguish between Model Coordinate System, World Coordinate System & Screen Coordinate System. 04
- (c)** Derive from fundamentals the necessary transformation matrix for rotating an object in anticlockwise direction by an angle θ about Y-axis. 04

SECTION - II

- Q4 (a)** What is Window? What is View Port? 03
- (b)** Give the steps for building up a computerized geometric solid model of a "Hexagonal Nut" using any solid modeling software. 04
- (c)** Describe the concept of parametric programming. Write a parametric program to draw a locating pin. 04
- Q5 (a)** Explain in brief different type of scanners. What are active scanner and passive scanner? 06
- (b)** The coordinates of four control points relative to a current WCS are given by $P_0 = [2 \ 2 \ 0]^T$, $P_1 = [2 \ 3 \ 0]^T$, $P_2 = [3 \ 3 \ 0]^T$ & $P_3 = [3 \ 2 \ 0]^T$. Find the equation of the resulting Bezier curve. Also find points on curve for $u = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, \text{ \& } 1$. 06

OR

- Q5 (a)** What is reverse engineering? Justify its use. 04
- (b)** List the various properties of Bezier curves 04
- (c)** A cubic Bezier curve is defined by points (1, 1) (2, 3) (4, 4) & (6, 1). Calculate the coordinates of parametric mid point of the curve & verify that its gradient is 1/7th at this point. 04
- Q6 (a)** Write a short notes on the following: 06
 1) DXF - Format
 2) STL - Format
- (b)** Develop an algorithm & interactive computer program for design of connecting rod of an IC engine. 06

OR

- Q6 (a)** Discuss the need for standardization in computer graphics & list the various CAD standards. 06
- (b)** For design of helical compression springs prepare an algorithm & interactive programme using any programming language. 06